

Z definice normy

$$\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2 = \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle - \langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle ,$$

díky linearitě (SL2) a skorosymetrii dostáváme

$$\langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \overline{\langle \mathbf{u}, \mathbf{v} \rangle} + \langle \mathbf{v}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \overline{\langle \mathbf{u}, \mathbf{v} \rangle} - \langle \mathbf{v}, \mathbf{v} \rangle = 2(\langle \mathbf{u}, \mathbf{v} \rangle + \overline{\langle \mathbf{u}, \mathbf{v} \rangle}) .$$

Protože $x + \bar{x} = 2 \operatorname{Re}(x)$, tak můžeme výraz dále upravit na $4 \operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle)$, z čehož po vydělení dostaneme zadanou rovnici.

Stejně vlastnosti skalárního součinu využijeme i při odvození druhé rovnice, navíc skutečnost, že $i(\bar{x} - x) = \operatorname{Im}(x)$.

$$\begin{aligned} \|\mathbf{u} - i\mathbf{v}\|^2 - \|\mathbf{u} + i\mathbf{v}\|^2 &= \langle \mathbf{u} - i\mathbf{v}, \mathbf{u} - i\mathbf{v} \rangle - \langle \mathbf{u} + i\mathbf{v}, \mathbf{u} + i\mathbf{v} \rangle = \\ &= \langle \mathbf{u}, \mathbf{u} \rangle - i\langle \mathbf{u}, \mathbf{v} \rangle + i\overline{\langle \mathbf{u}, \mathbf{v} \rangle} - \langle \mathbf{v}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{u} \rangle - i\langle \mathbf{u}, \mathbf{v} \rangle + i\overline{\langle \mathbf{u}, \mathbf{v} \rangle} + \langle \mathbf{v}, \mathbf{v} \rangle = 2i(\overline{\langle \mathbf{u}, \mathbf{v} \rangle} - \langle \mathbf{u}, \mathbf{v} \rangle) = \\ &= 4 \operatorname{Im}(\langle \mathbf{u}, \mathbf{v} \rangle) \end{aligned}$$